

Unsupervised Learning by Neural Networks: a light introduction

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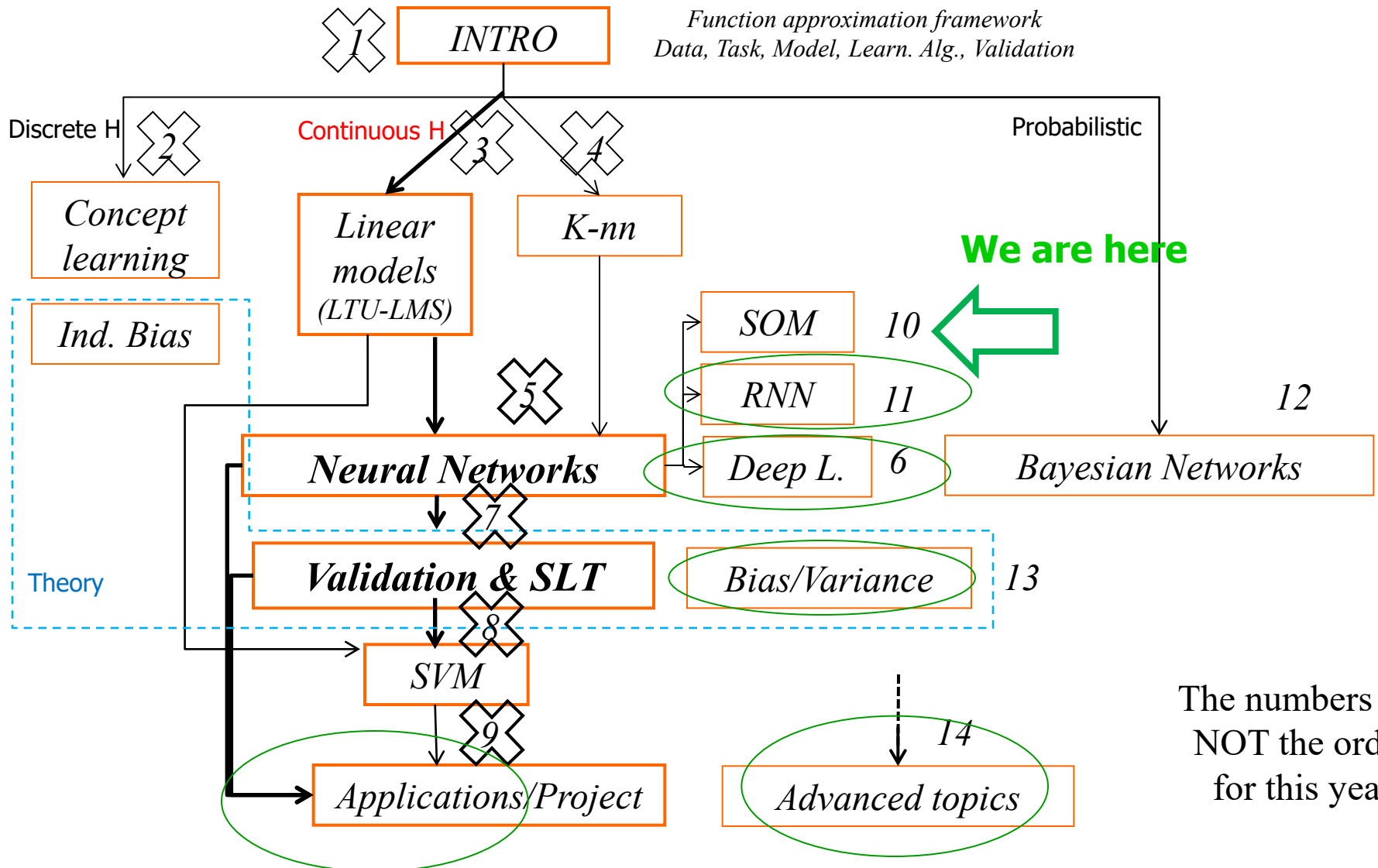
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ML Course structure

Where we are



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Tasks: Unsupervised Learning



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Unsupervised Learning: No teacher!

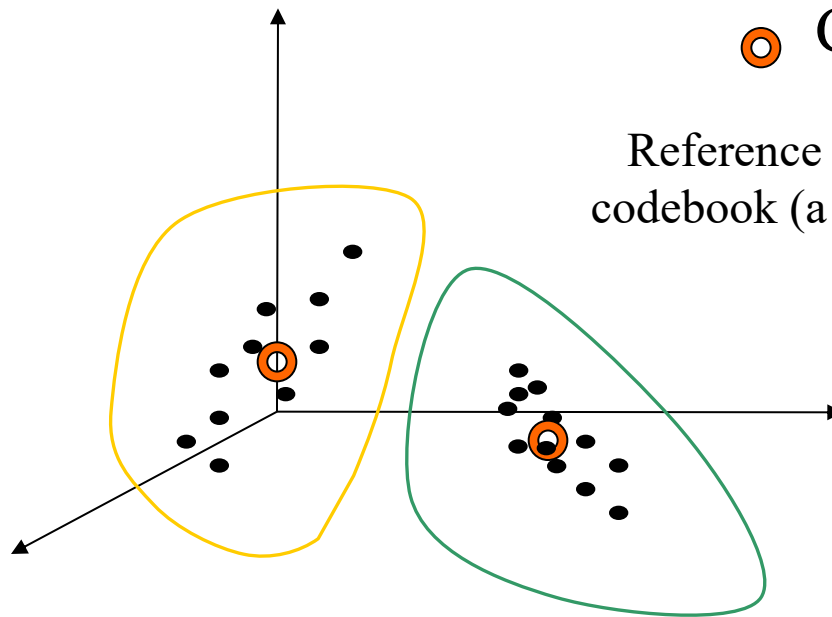
- TR = set of unlabeled data $\langle \mathbf{x} \rangle$
 - Clustering: find *natural groupings* in a set of data
 - Dimensionality reduction/ Visualization/Preprocessing:
multidimensional data are projected in a lower dimensional space
 - Principal Component Analyses, Multi-Dimensional Scaling, Independent Component Analysis, ...
 - Modeling the data density (not in this course)

Utility

- Example (Taxonomy): organize life forms: what are mammals ?
- Example (human learning): babies learn to recognize ("cluster") familiar faces before they can understand human speech
- Unlabeled data is much cheaper to obtain
 - Examples: e.g. 10 million images (sampled frames from YouTube) with state-of-the-art results on object recognition (22,000 categories)

Clustering

Partition of data into clusters (subsets of “similar” data):
patterns within a valid cluster are more similar to each other
than they are to a pattern belonging to a different cluster



○ Centroids (or prototypes)

Reference vector, centroid, cluster center,
codebook (a set of codevectors) , or prototype
vector, ...

A guideline

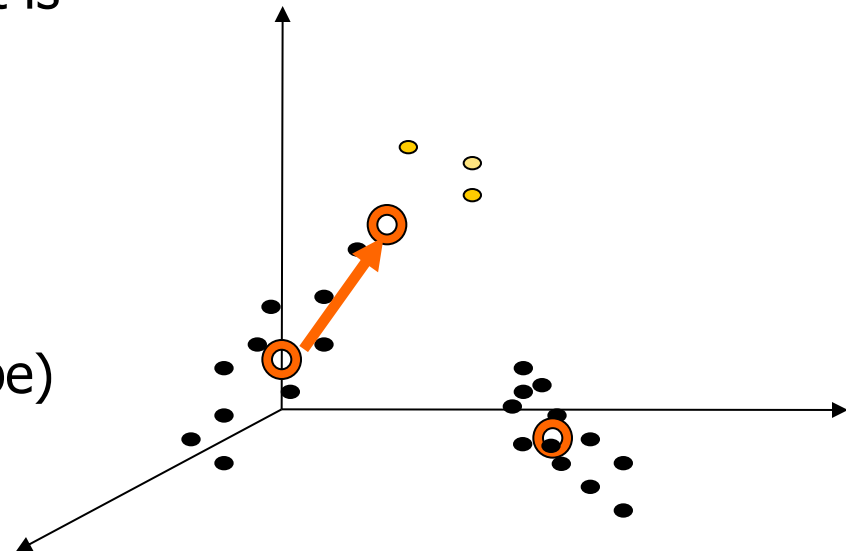
- We mainly focus on Clustering/Vector Quantization
- The clustering problem has been addressed in many contexts and by researchers in many disciplines;
- *Thousand of different algorithms !*
 - this reflects its broad appeal and usefulness as one of the steps in exploratory data analysis.
 - **QUESTION:** how many of you already know the “K-means” ?
- We don’t try to list them, but we go directly to a simple approach and we have the opportunity to look at a historical NN models:
 - the **SOM/Kohonen maps !**
- We use a Vector Quantization perspective
- for Partitioning clustering

A premise: Similarity Measures



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- Metric dominance ! The assumed distance tell us which are the similar examples (Pattern proximity)
- The clustering algorithm decides the grouping basing on that metric
- There are many possible metrics
- Although Euclidian metric is popular it is just a choice !
- *See Nearest Neighborhood lecture !*
- May have more deep implications:
Soft outlier weight by using
(1-Gaussian centered in the prototype)



Vector Quantization

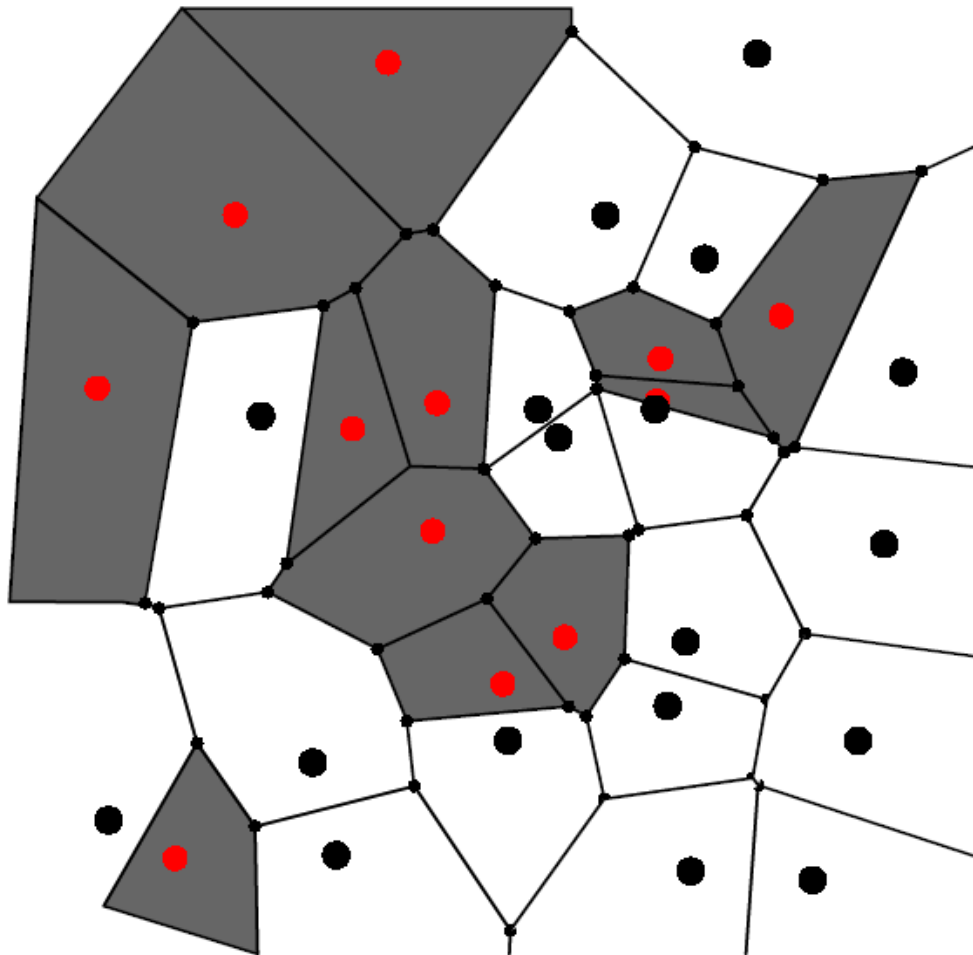
- Vector quantization techniques encode a data manifold, e.g., a submanifold $V \subseteq R^n$, utilizing only a finite set $\mathbf{w} = (\mathbf{w}_1, \dots, \mathbf{w}_K)$ of reference or “codebook” vectors $\mathbf{w}_i \in R^n$, $i = 1, \dots, K$.

- A data vector $\mathbf{x} \in V$ is described by the best-matching or “winning” reference vector $\mathbf{w}_{i^*(\mathbf{x})}$ of $\mathbf{w}$ for which the *distortion error* $d(\mathbf{x}, \mathbf{w}_{i^*(\mathbf{x})})$ is minimal
- This procedure divides the manifold V into a number of subregions

$$V_i = \left\{ \mathbf{x} \in V \text{ s.t. } \|\mathbf{x} - \mathbf{w}_i\| \leq \|\mathbf{x} - \mathbf{w}_j\| \forall j \right\}$$

- called Voronoi polyhedra, out of which each data vector $\mathbf{x}$ is described by the corresponding reference vector $\mathbf{w}_{i^*(\mathbf{x})}$

Voronoi diagram



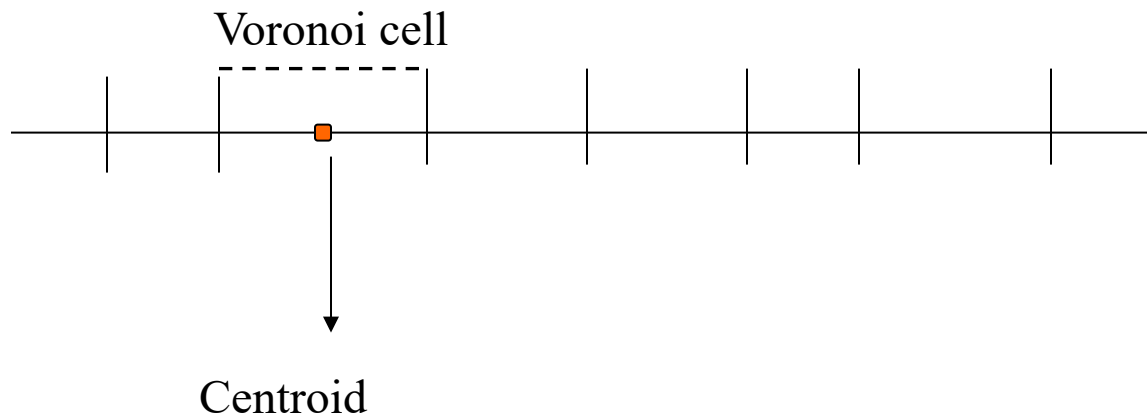
Each cell consisting of all points closer to *the bold black/red points* than to any other patterns.

The segments of the Voronoi diagram are all the points in the plane that are equidistant to two patterns.

NOTE: for us the problem is: **given the dataset, find centers such that MIN Loss E**

In general Hard to be computed:
Optimal vector quantization codebook is an NP-complete problem

Example: Quantization in 1D



- E.g. In digital signal processing, **quantization** is the process of approximating ("mapping") a continuous range of values (or a very large set of possible discrete values) by a relatively small ("finite") set of ("values which can still take on continuous range") discrete symbols or integer values
 - so we have a quantization/distortion error
- *Music in your digital device !!!!*

VQ and Clustering

- Note that clustering has a more general objective of finding *interesting/useful* groupings of data
- *Interesting* is often implicitly defined via the computational procedure by itself and not necessarily measured by the minimum Quantization error
- However, VQ provides an useful framework, and VQ algorithms can be (and are often) used for clustering and we use this nice perspective

Clustering and Vector Quantization



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- Goal: optimal partitioning of unknown distribution in x -space into regions (clusters) approximated by a cluster center or *prototype*.
- I.e a set of vector quantizers $x \rightarrow c(x)$ or $w_{i^*(x)}$
continuous space $\rightarrow$ discrete space (the number of centers)
- considering the squared error distortion
- We evaluate $d(x_i, c(x_i)) = \|x_i - c(x_i)\|^2$
- The average value over the distribution of inputs is the average (expected) distortion or reconstruction or quantization error.
This is our *Loss function*
- We use w as symbol for the prototype vectors

Quantization error function

$$E = \int f(d(\mathbf{x}, \mathbf{w}_{i^*(\mathbf{x})})) p(\mathbf{x}) d\mathbf{x} = \int \|\mathbf{x} - \mathbf{w}_{i^*(\mathbf{x})}\|^2 p(\mathbf{x}) d\mathbf{x}$$

(squared error criterion)

$$E = \sum_i^l \sum_j^K \|\mathbf{x}_i - \mathbf{w}_j\|^2 \delta_{winner}(i, j) \quad \text{Discrete version}$$

$\delta_{winner}(i, j)$ characteristic function of the receptive field of $\mathbf{w}_j$
1 if j is the winner for $\mathbf{x}_i$, 0 otherwise

$p(\mathbf{x})$ is the probability distribution of $\mathbf{x}$

Minimize E

$$E = \int f(d(\mathbf{x}, \mathbf{w}_{i^*(\mathbf{x})})) p(\mathbf{x}) d\mathbf{x} = \int \|\mathbf{x} - \mathbf{w}_{i^*(\mathbf{x})}\|^2 p(\mathbf{x}) d\mathbf{x}$$

Minimizing E (in $\mathbf{w}$) $\rightarrow$ vector quantization

- Minimize E through an optimal choice of $\mathbf{w}$
- The **set of reference vectors $\mathbf{w}$** that minimizes E **is the solution** of the VQ problem
- *Note:* the integrand of E is not continuously differentiable:
 - Indeed $\mathbf{w}_{i^*(\mathbf{x})}$ have **discrete** changes (w.r.t. $\mathbf{x}$).
 - However, we compute the derivatives locally (fixing $\mathbf{x}$ on a Voronoi cell)
 - ...

On-line K-means

$$E = \sum_i \sum_j \|\mathbf{x}_i - \mathbf{w}_j\|^2 \delta_{winner}(i, j)$$

- Taking the derivatives of E with respect to the $\mathbf{w}_j$ yields the learning rule of VQ
- Lloyd and MacQueen's well-known **K-means** clustering algorithm (on-line version) for any $\mathbf{x}_i$

$$\Delta \mathbf{w}_{i^*} = \underset{\substack{\uparrow \\ \text{Learning rate}}}{\eta} \delta_{winner}(i, i^*) (\mathbf{x}_i - \mathbf{w}_{i^*})$$

- Note that we change only the winner prototypes, adapting it to $\mathbf{x}$
 - A stochastic gradient descent $\rightarrow$ local minima
 - Method dependent on initialization
 - Moreover, the number of clusters to find must be provided

Basic K-means Batch Alg.



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LBG(Linde, Buzo, Gray) or generalized Lloyd algorithm

- (1) Choose k cluster centers to coincide with k randomly-chosen patterns or k randomly defined points inside the hypervolume containing the pattern set.
- (2) Assign each pattern to the closest cluster center (the winner)
- (3) Recompute the cluster centers (geometric centroid, i.e. **mean**) using the current cluster memberships
- (4) If a convergence criterion is not met, go to step 2. E.g.
 - no (or minimal) reassignment of patterns to new cluster centers,
 - minimal decrease in squared error.

Exercise: from on-line to batch alg: sum (mean) over patterns the on-line delta (without eta, i.e. eta=1) and apply to obtain $\mathbf{w}_{\text{new}} (= \mathbf{w}_{\text{old}} + \dots)$

Details

- Given the cluster centers $w_1 \dots w_K$ (vectors of dim n , as $\mathbf{x}$)
- (2) For each $\mathbf{x}$ the winner is (the nearest prototype):

$$i^*(\mathbf{x}) = \arg \min_i \|\mathbf{x} - w_i\|^2 \rightarrow \text{Norm 2}$$

$\rightarrow \text{find } i \text{ s.t. dist. is min}$

Euclidian distance !

$$\|\mathbf{x} - w_i\|^2 = \sum_{j \rightarrow n} (x_j - w_{ij})^2$$

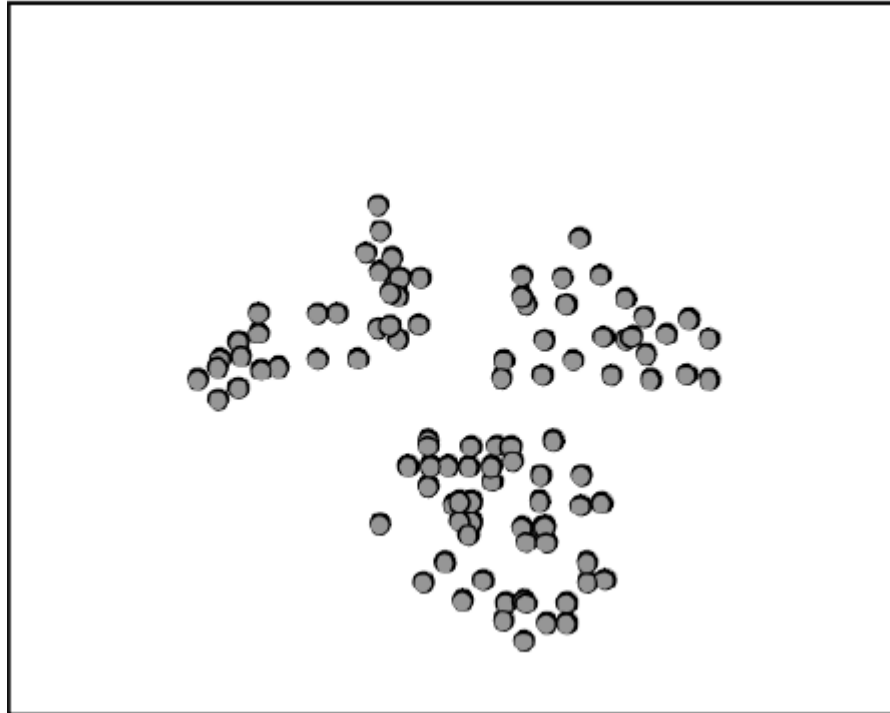
- Now $\mathbf{x}$ belong to cluster i^*
- (3) For each cluster i the new mean (centroid) is:

$$w_i = \frac{1}{|\text{cluster}_i|} \sum_{j: x_j \in \text{cluster}_i} \mathbf{x}_j$$

$\swarrow$ Vector $\searrow$ Vectors
 $\quad \quad \quad \searrow$ Number of cluster members

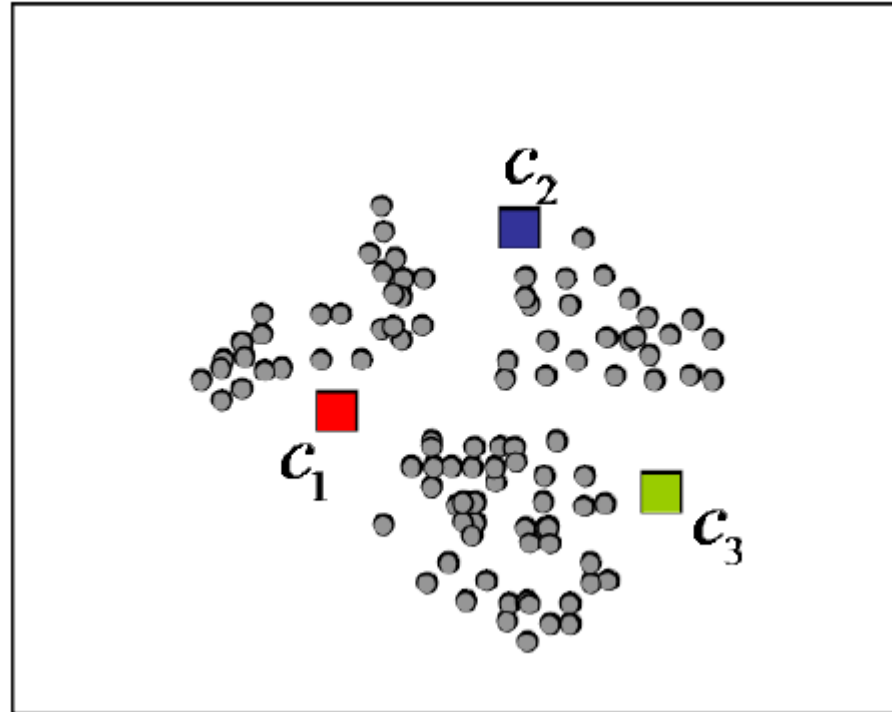
*Note: $\|\cdot\|^2$ is the square of norm 2,
hence the sum of squares (no root)*

K-means example (1)



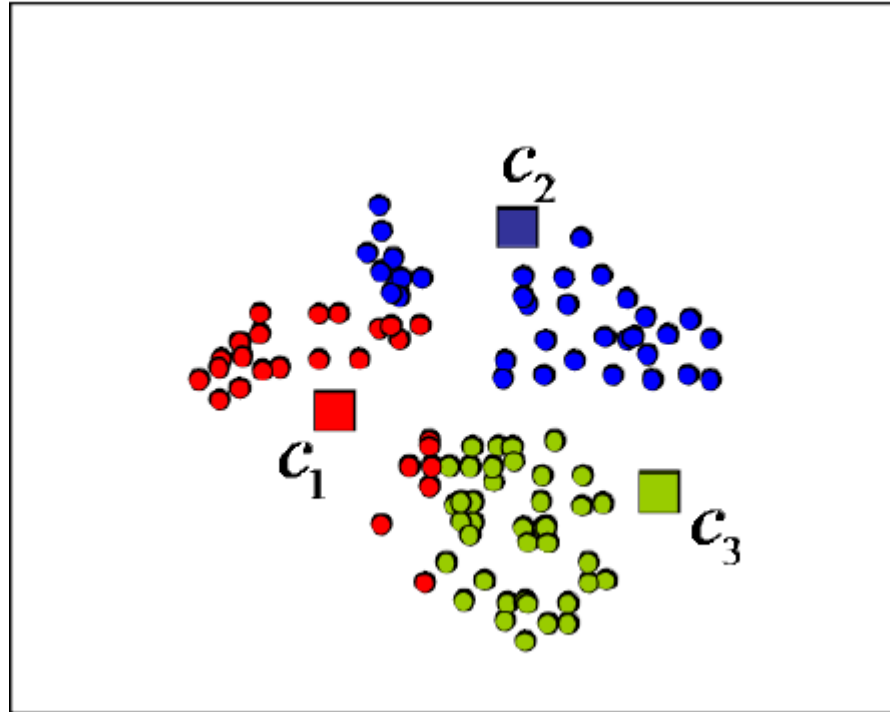
Input data

K-means example (2)



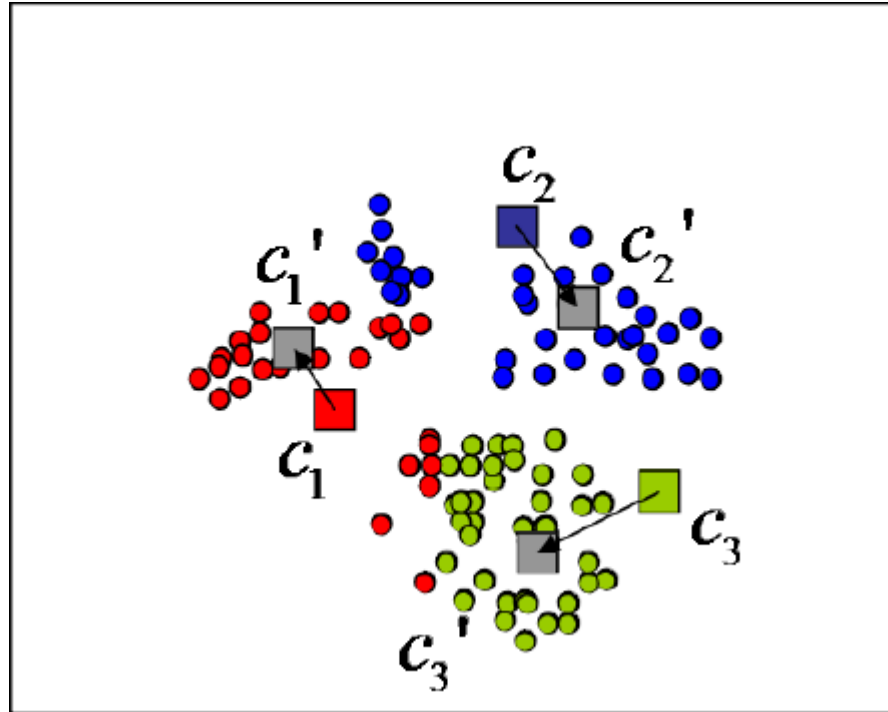
Initialize $K = 3$ prototypes

K-means example (3)



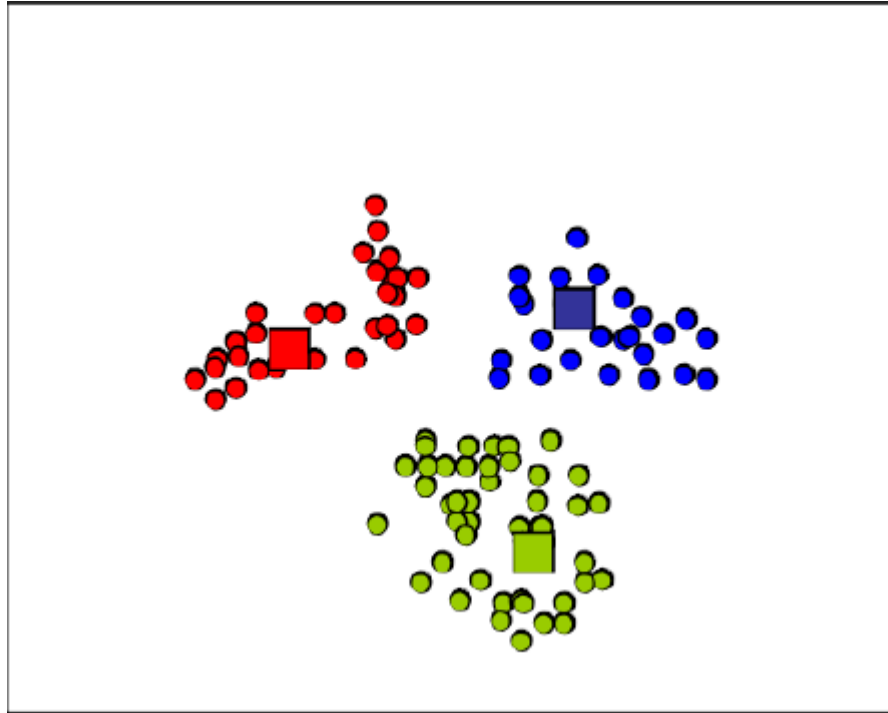
Assign samples to the nearest
prototype/cluster

K-means example (4)



Compute the updated prototypes C'

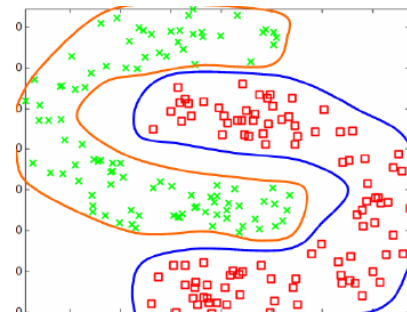
K-means example (5)



Update cluster assignment

K-means (+/-)

- The k -means is the simplest and most commonly used algorithm employing a squared error criterion
- The k -means algorithm is popular because it is easy to implement, and its in general efficient
- The number K of clusters to find must be provided
- Local minima of E make the method dependent on initialization and only sub optimal choice of the reference vectors !
- It can work very well for compact and hyperspherical clusters.
- No visualization properties !
 - K-means does not allow us to project data in a lower dim space
 - Note that the indexing of these w can be made in an arbitrary way, whereby this mapping is still *unordered*.



NOT
hyperspherical
clusters:
what happens?

Soft-max

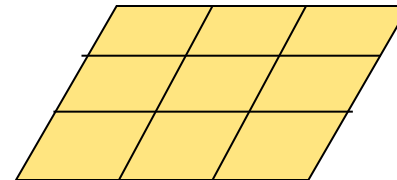
- To avoid confinement to local minima, a common approach is to introduce "soft-max" adaptation rule I.e.
- Not only the winning reference is modified
- The adaptation to x affects all cluster centers depending on their proximity to x , typically with a step size that decreases with the distance $d(x, w_i)$
- Maximum-entropy clustering (using Gaussians distance)
- An instance of this strategy in NN: the **Kohonen self-organizing maps**, where the proximity among reference vectors is defined on a ordered map, I.e. a neural grid where references vectors are arranged (and associated to units)

Unsupervised learning in NN: e.g. SOM



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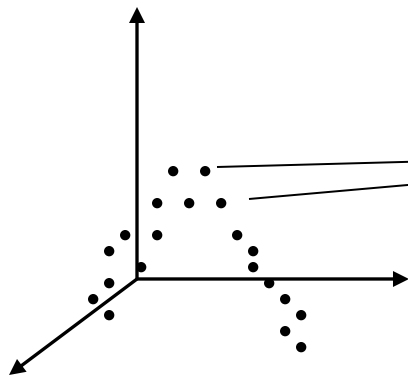
- **Self-organizing map (SOM)** neural networks, also called **Kohonen maps**, from the author's name (1981)
- Consist of N neurons located on a regular low-dimensional, usually 2-D, grid (lattice).
 - Each unit can be identified by the coordinate on the map
 - Each unit receives the same input x
 - Each unit has a weight w



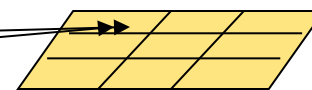
Neural map (2D grid): 3 X 3 neurons

Unsupervised learning in NN: SOM overview (AIM)

- SOM: learn a map from input space to a lattice of neural-units
 - topology-preserving map:
 - neighboring units respond to similar input pattern
 - data points close in the input space are mapped onto the same or **nearby map units***
 - visualization properties of the SOM output feature map (2D)



3-D (or multiDim. in general)



Neural map (2D grid)

The unit whose weight vector is closest to the input is the **winner**.

SOM overview: Use of SOM

- **Clustering:** clusters can be identified on the map
(rapidly find structures underlying the data)
- **Pattern recognition and vector quantization:** the input is transformed into the closest neuron weights, i.e. the reference or "codebook" vector;
- **Data compression:** the input is transformed into the indexes, or digital codes, of the winner unit;
- **Projection and exploration:** the distribution of multidimensional data is visualized in a lower dimensional space (the 2D map).

SOM overview: Competitive Learning



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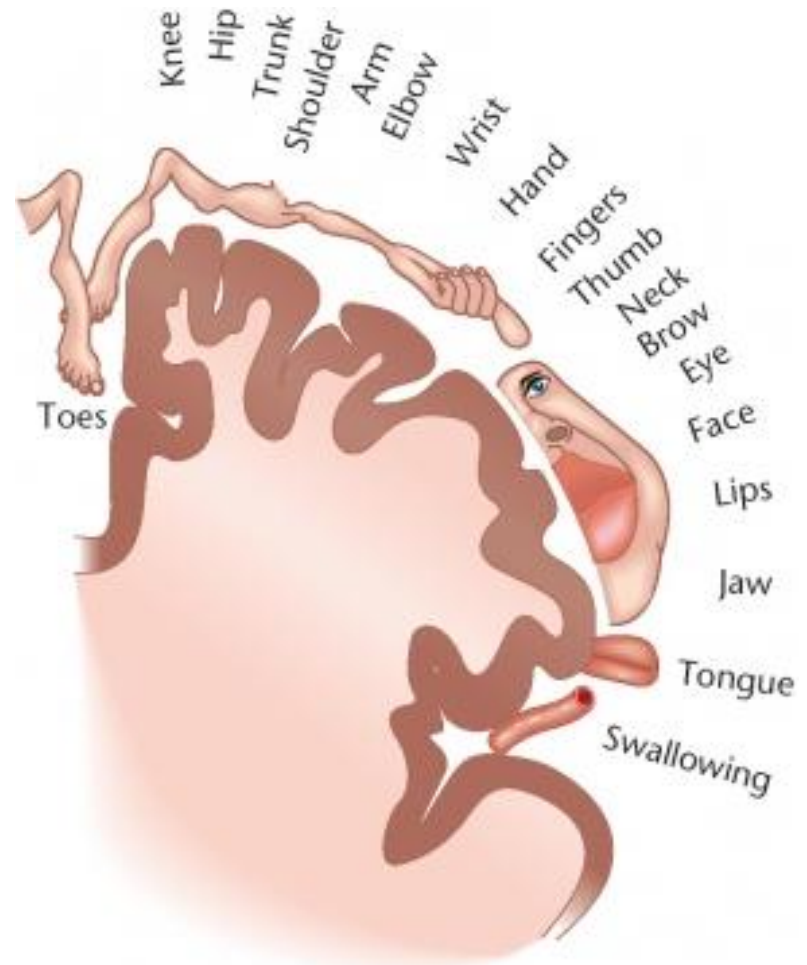
- **Competitive Learning** is an adaptive process in which neurons gradually become sensitive/specialized to different input categories/set of samples
- Competitions among neurons for an input data → the winner is allowed to learn more.

SOM overview: Homunculus in our brain

Somatosensory cortex (Motor cortex is similar)

Topologically ordered (feature) map

- **Adjacent neurons** come to represent coincident source of activations (in the body)
- Larger area of cortex for more sensitive body parts
- In a **topology-preserving map**, units located physically next to each other will respond to classes of input vectors that are likewise next to each other.
- The nature (biology) exploited this mechanism!

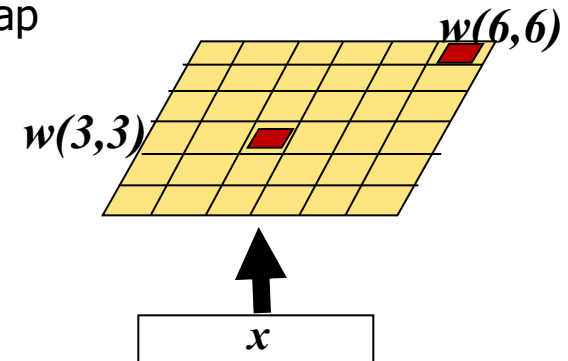


SOM model and algorithm

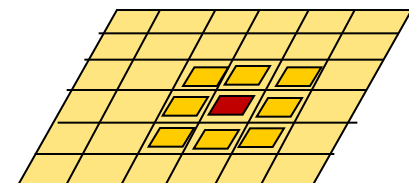
- The aim of SOM algorithm is to learn a map from vectors to a discrete output space (the coordinates of the unit on the map)
 - Each unit can be identified by the coordinate on the map
 - Each unit receives the same input x
 - Each unit has a weight w
 - $\text{Dim}(w) = \text{dim}(x) = n$

Learning Alg.

- The map is randomly initialized (weight values)
- Drawn a sample input x
- **Competitive stage:** the winner is the unit (with w) most similar to x
- **Cooperative stage:** upgrade the weights of the units that have topological relationships (on the map) with the winner unit (soft-max).
- Continues until convergence (no changes)



KEY FACTOR: neighbors on the map!!!!
(instead, standard soft-max used proximity in the data space)



SOM Learning Alg. (1)

Competitive stage

The current input vector $\mathbf{x}$ is compared with all the unit weights using an *Euclidean* distance criteria.

Standard SOM adopts a “winner-take-all” strategy
(we already seen this in the K-means):

The winner is

$$i^*(\mathbf{x}) = \arg \min_i \|\mathbf{x} - \mathbf{w}_i\|$$

where i is an index on the map (e.g. coordinate in 2D)

SOM Learning Alg. (2)

Cooperative stage

- The weight of the winning unit and the weight of the units in its neighborhood are moved closer to the input vector (*Hebbian learning*)

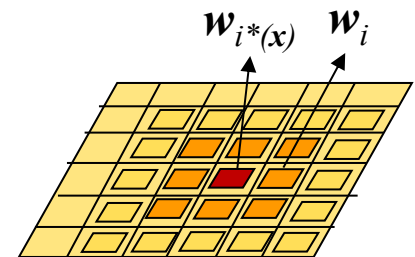
- At iteration t
$$\mathbf{w}_i(t+1) = \mathbf{w}_i(t) + \eta(t) h_{i,i^*}(t) [\mathbf{x} - \mathbf{w}_i(t)]$$

- where $h(t)$ is the neighborhood function (or neighborhood kernel), i.e. $h_{i,j}(t)$ is a function that decreases monotonically for increasing $\|i-j\|$.

$$h_{i,i^*}(t) = \exp \left(- \frac{\|\mathbf{r}_i - \mathbf{r}_{i^*}\|^2}{2\sigma_{nh}^2(t)} \right)$$

Coord. of unit i

width



- Learning rate and neighbourhood radius values (σ) are decreased as function of iteration t
 - neighbourhood wide at the beginning (global ordering) and then width and height slowly decrease during learning (convergence)
 - In practice, the shape of the neighborhood is chosen from standard geometrical form (rectangular, hexagonal etc. on the SOM grid) to include a finite set of the map units.

The Emerging Topological Order



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- It is important to stress that the update rule of the cooperative stage is fundamental to the formation of topographically ordered maps.
- In fact, the weights are not modified independently of each other but as topologically related subsets.
 - For each subset similar kinds of weight updates are performed.
 - For each step a subset is selected on the basis of the neighborhood of the current winner unit (spatially, in the lattice).
 - Hence, topological information is supplied to the map because both the winning unit and its lattice neighbors receive similar weight updates that allow them, after learning, to respond to similar inputs.
- Although this seems quite natural, a formal proofs of ordering is difficult (see Kohonen book in the bibl.)
- Exercise: focalize the differences between K-means and SOM approaches.

SOM training on a 2D uniform data distribution

2D map, similar reference vectors (weights) are connected by lines

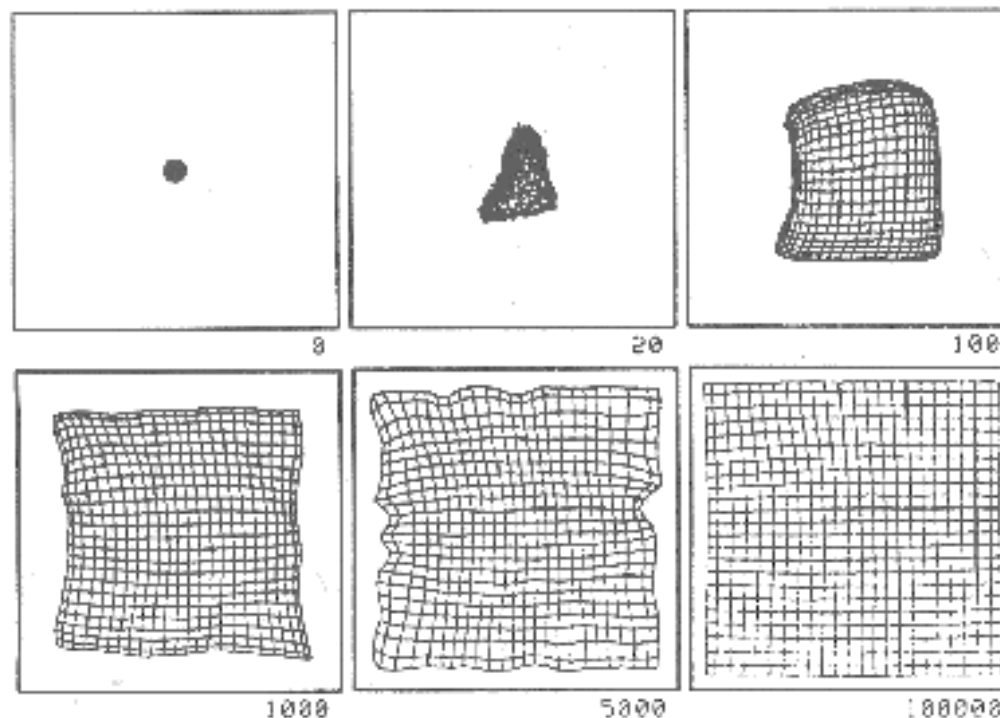
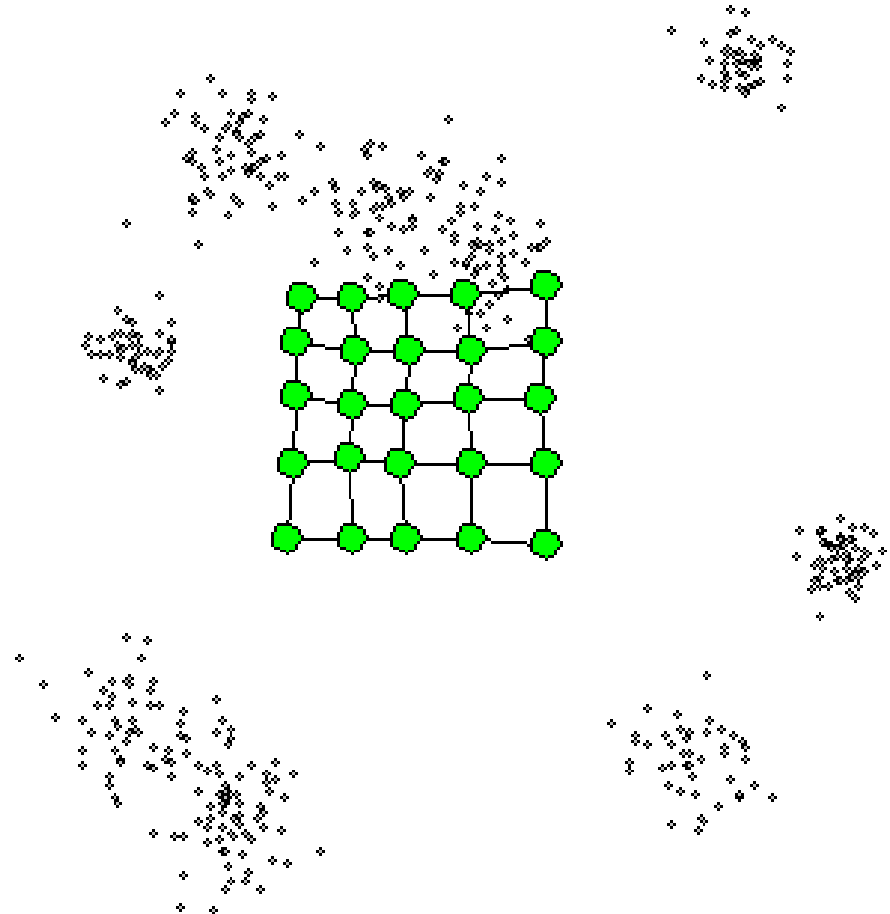


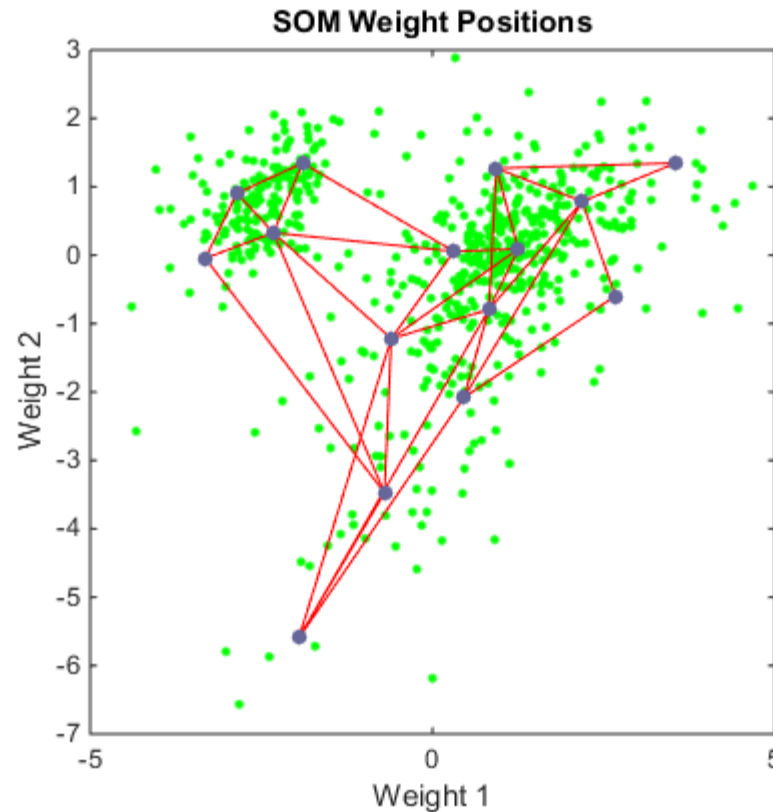
Fig. 3.4. Reference vectors during the ordering process, square array. The numbers at lower right-hand corner indicate learning cycles

Training the SOM (animated gif)

Units “go to” represent the data clusters



Static image (after training on a different data set)



But for another movie, e.g.:

<https://www.youtube.com/watch?v=k7DK5fnJH94>

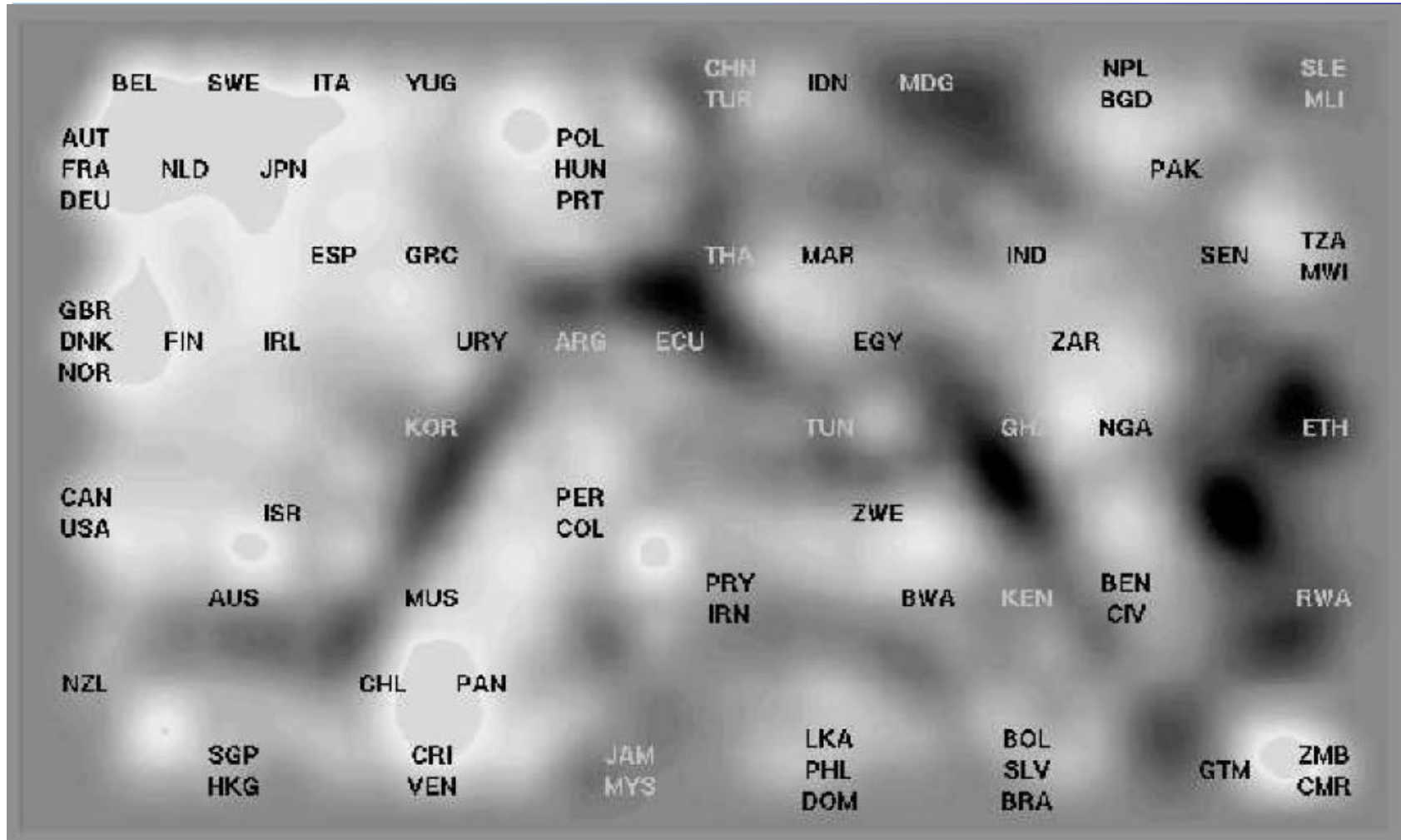
Visualization of SOM

- The map can be used to visualize different features of the SOM and of the data represented on the map. E.g.
 - **density** of the reference vectors (color density proportional to hits)
 - **distance** between prototype vectors of neighboring units (U-matrix):
where dark color are used to indicate large distance between adjacent map units (prototypes) and light color are used to indicate small distances between adjacent map units. Next example.
 - Sammon maps (using lines)...
- Providing the SOM with easy interpretability!

An Applicative Example

- Data set of welfare indicators of different countries used to investigate the economic relationships between their economic systems.
- 39-Dim data (Welfare descriptors), 77 countries, according to World Development Report 1992
- On the map display, the grey level of the point that is in the middle of two map units is determined by the scaled distance value between the pair of corresponding reference vectors (w) and then spatially smoothed. (light color are used to indicate small distances between adjacent map units)
- S. Kaski, T. Kohonen. Exploratory data analysis by the self-organizing map: Structures of welfare and poverty in the world. 1995

An Applicative Example



A smoothed distance matrix of 13 X 9 SOM lattice (illustrating the clustering structure)

Note that there is no need for a priori assumption on the shapes of the clusters

Evaluation (just few words)

- How is the output of a clustering algorithm evaluated?
- What characterizes a 'good' clustering result and a 'poor' one?
- Subjective: little in the way of 'gold standards' exist in clustering except in well-prescribed subdomains.
- Objective measures (no treated here): e.g. quantization error (but note that depends on the number of centroids)!
- But not "trivially" the label of a underlining classification task, called ***purity*** (entropy, mutual inf., ...) (else you can use a classifier)!
- Quality of model/clustering versus quality of results (domain dependent)

Other topics in Unsupervised

- Hierarchical Clustering → *dendrograms*
- Principal Components, Curves and Surfaces Analysis
- Association Rules [DM]
- Independent Component Analysis and Exploratory Projection Pursuit
- Multidimensional Scaling
- t-distributed stochastic neighbor embedding (t-SNE)
- Generative Topographic Mapping (**GTM**)
- **ART** (adaptive resonance theory) NN
- ...

Exercises

- K-means: See before from on-line to batch alg.
- Resume the difference between SOM and K-means

Bibliography

Basic

- Haykin (2nd ed.): Chapter 9 (up to 9.6)
- Haykin (3rd ed.): Chapter 9 (up to 9.5/9.6 – thought 9.6 is new)
- Duda, Hart, Stork: Chapter 10 (selecting section)

Many others:

- Many Tutorials on Journals
- E.g. for SOM
- (a book) Kohonen 1995 (2nd ed. 1997) Self-organizing Maps, Springer
- An impressive number of applications papers!
- There is a paper collecting ref. of 3343 scientific papers on SOM up to 1997 [Bibliography of self-organizing map (SOM) papers: 1981-1997].... On 2015 probably more than 8000, and nowadays?

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